

Mini Shell – User Documentation

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Course: COD 7001 (System Concepts)

1 Problem Statement

Design and implement a UNIX-style command-line shell capable of executing external programs, managing processes, handling I/O redirection, pipelines, and basic job control, without invoking another shell.

2 Introduction

The Mini Shell is a simplified UNIX-like command-line interpreter demonstrating concepts such as:

- Process creation using `fork()` and `execvp()`
- Input/output redirection
- Command parsing
- Pipelines
- Background execution
- Signal handling

The shell displays the current working directory as its prompt and waits for user commands in an interactive loop.

3 Starting the Shell

Compilation

```
make
```

Running

```
./mini-shell
```

Prompt Example

```
/home/hobbbbit31/Desktop/mini-shell $
```

4 Feature Overview

The Mini Shell supports:

- Executing external commands
- Built-in commands (`cd`, `pwd`, `exit`, `history`)
- Background execution using `&`
- Input/output redirection
- Quoted strings
- Single pipeline using `|`
- Ctrl+C handling without shell termination
- Zombie process cleanup

5 Executing External Commands

Any program available in the system PATH can be executed.

Examples:

```
ls -l
echo Hello
cat notes.txt
```

6 Built-in Commands

6.1 cd – Change Directory

```
cd <directory>
cd ~
cd
```

6.2 pwd – Print Working Directory

```
pwd
```

6.3 exit – Quit the Shell

```
exit
```

7 Background Execution (&)

Running a command in the background:

```
sleep 5 &
```

8 Input/Output Redirection

8.1 Input Redirection (i)

```
grep error < log.txt
```

8.2 Output Redirection (i)

```
ls > files.txt
```

8.3 Append Output (i)

```
echo done >> progress.log
```

9 Quoted Strings

Examples:

```
echo "Hello World"
grep 'user not found' auth.log
```

10 Pipelines

```
ls | wc -l
```

11 Ctrl+C Handling

Ctrl+C interrupts only the running child process. The shell remains active.

12 Zombie Process Cleanup

A SIGCHLD handler automatically reaps background processes:

```
signal(SIGCHLD, handle_zombie);  
waitpid(-1, NULL, WNOHANG);
```

13 Error Handling

The shell reports:

- Missing filenames after redirection
- Misuse of &
- Multiple redirections
- Incorrect pipeline structure
- Unmatched quotes

14 Usage Examples

14.1 Simple

```
ls -l  
echo "Mini Shell"
```

14.2 Redirection

```
sort nums.txt > sorted.txt  
grep hello < input.txt >> output.log
```

14.3 Pipelines

```
ls | wc -l
```

14.4 Combined

```
cat input.txt | grep pass > result.txt &
```

15 Exiting the Shell

```
exit
```